Python CGPA Calculator

Overview

This project is a simple, command-line Python program that helps students calculate their CGPA on a 10-point scale. Users just need to enter the marks and course credits for each subject, and the program takes care of the rest.

It checks the inputs, converts marks into grade points, and then computes the final CGPA. This tool is ideal for students or colleges that want a quick and effortless way to track academic performance.

☐

Features

- Converts percentage marks into grade points

(10 for 90+, 9 for 80–89, 8 for 70–79, etc.)

- Works for any number of subjects or courses
- Ensures marks are within 0–100 and credits are positive
- Handles wrong or non-numeric inputs gracefully by skipping invalid subjects
- Displays the final CGPA in a clear and user-friendly format
- Notifies the user if no valid subjects or credits were entered

☐

Technologies Used

- Python 3 (tested with Python 3.6 and above)
- No external libraries required — uses only standard input and print

Steps to Install & Run the Project

Make sure Python 3 is installed in your machine.

Save the source code as cgpa_calculator.py.

Open the terminal and cd into the folder containing cgpa_calculator.py.

Run the program using:

Text

```
python cgpa_calculator.py
```

Follow the prompts: Enter the number of subjects, marks (0–100), and credits for each subject.

Instructions for Testing

Test normal cases, ensuring valid marks and positive credits. Verify correct computation of CGPA.

Test invalid marks (< 0, > 100) to test error handling. Test invalid credits for validation: zero, negative. Test non-numeric input for marks/credits to verify user prompt and skipping behavior. Verify result if zero valid subjects are entered. Should display that CGPA cannot be calculated. Screenshots (Optional but Recommended) Screenshot of a typical run: user provides subjects and gets a result for CGPA. Screenshot showing the error message for invalid marks. Screenshot showing the program skipping a subject for non-numeric input.